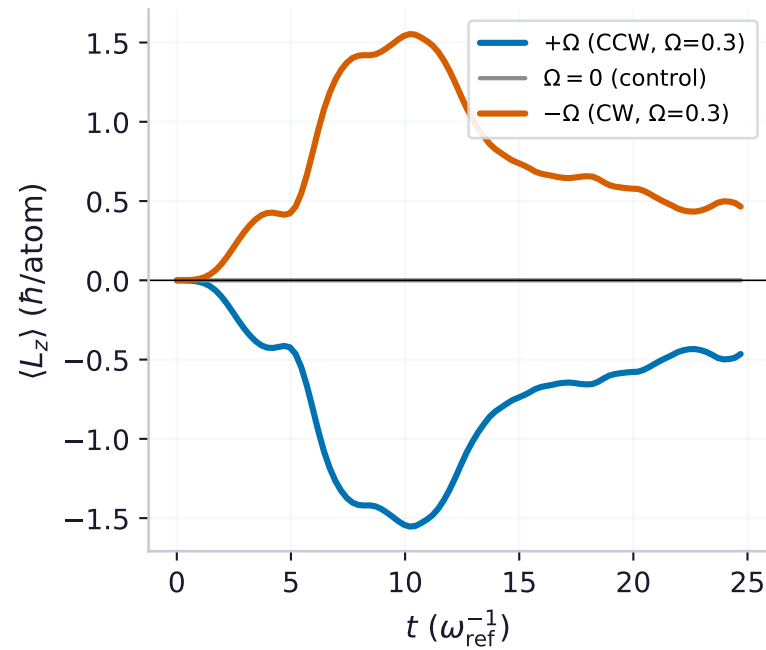
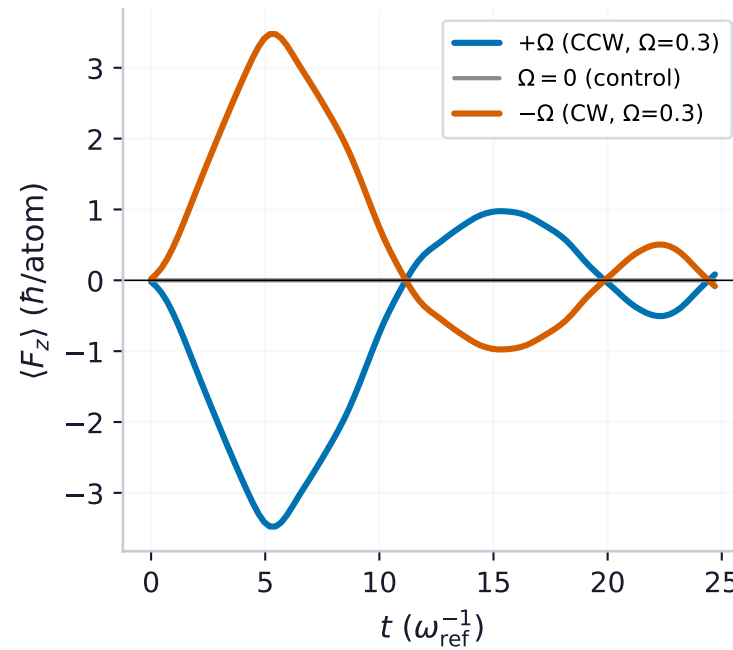


Rotation direction controls vortex chirality + Barnett spin sign (^{151}Eu $F=6$ dipolar BEC, optimal $\Omega=0.30$)

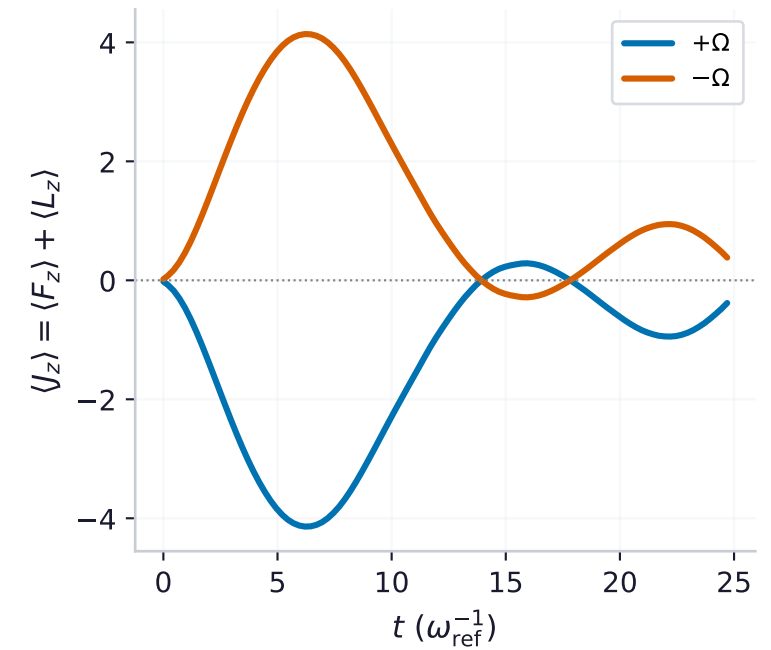
(a) orbital angular momentum (vortices)



(b) Barnett axial magnetisation

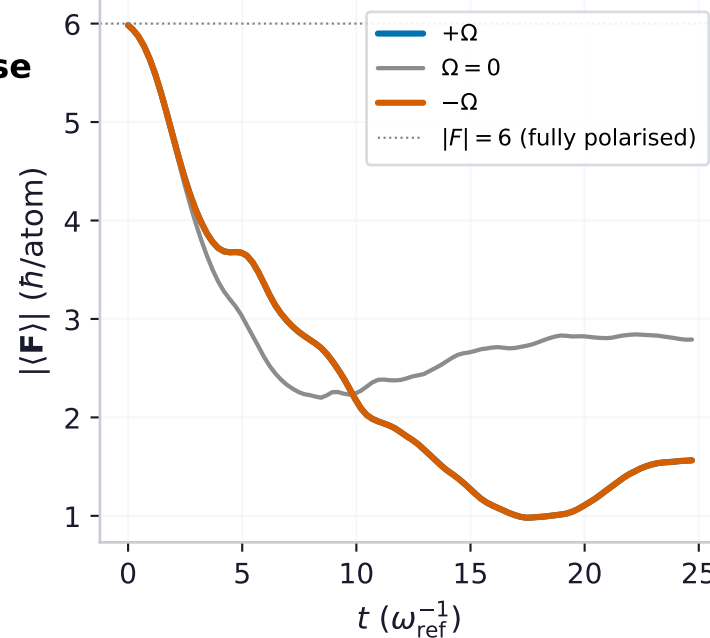
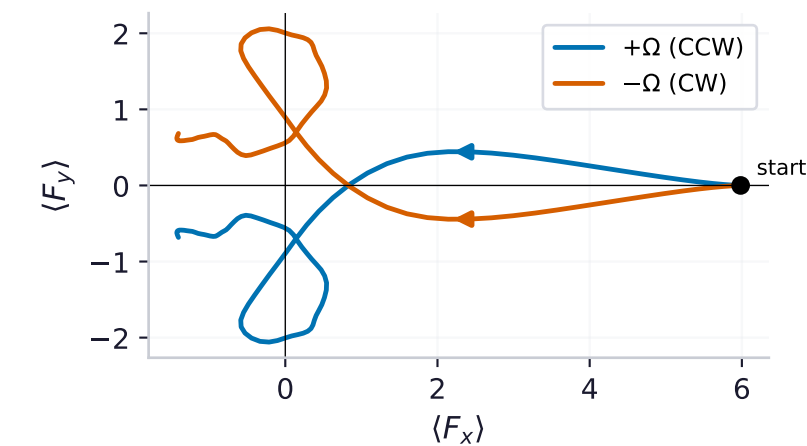


(c) J_z pumped by the drive (mirror-antisymmetric)



(e) genuine depolarisation (many-body, not Larmor)

(d) transverse spin winds with rotation sense



Rotation-direction control
(transverse $J_z=0$ start, unitary, $\Omega=0.3$)

peak $|\langle L_z \rangle| = 1.55 \hbar/\text{atom}$
peak $|\langle F_z \rangle| = 3.48 \hbar/\text{atom}$

EXACT MIRROR under $\Omega \rightarrow -\Omega$:

$\max |L_z(+\Omega) + L_z(-\Omega)| = 0.0e+00$

$\max |F_z(+\Omega) + F_z(-\Omega)| = 0.0e+00$

(residual exactly 0: the seed-free run is symmetric under $\Omega \rightarrow -\Omega$, $y \rightarrow -y$, so the $-\Omega$ run IS the y -reflection of $+\Omega$)

$\Omega=0$ control: $|\langle L_z \rangle|_{\text{max}} = 0.0e+00$

$\rightarrow$ no rotation, no vortex, no Barnett
(spin still depolarises via DDI: $|F|$ falls, but with zero net L_z , F_z).

Left (CW, $-\Omega$) and right (CCW, $+\Omega$) rotation give opposite-sign vortices and opposite-sign Barnett magnetisation.